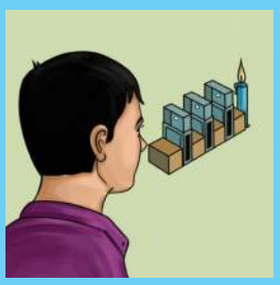
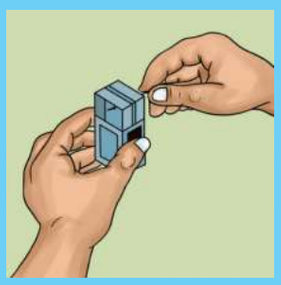
ACTIVITY 1

Requirement: Three empty match boxes, pin, candle and wooden blocks.





Procedure: Arrange empty match boxes and wooden blocks as shown in the figure. First, you make a hole in the inner tray of each match box such that all three holes are in the same spot. Arrange the match boxes as shown in figure. Now, adjust the three inner trays in such a way that the three holes are in a straight line. Place a lighted candle at one end of this arrangement and try to see the flame of candle from a hole at the other end. Is the flame visible?

Now, arrange the trays such a way that they are not at the same height. Try to see the flame. Is it visible? What does this activity tell you about the path of light?

Light travels in straight line, it cannot bend the path itself. This is called as the *rectilinear propagation of light*. This is one of the most important property of light.

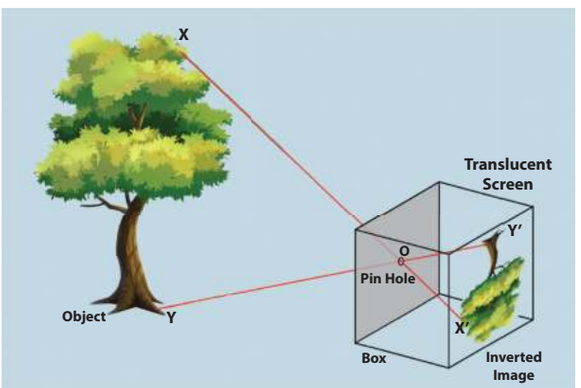


Al-hasan-Haytham was a scientific thinker who made important contribution to the understanding of vision, optics and light. He observed that light coming through a tiny hole travelled in straight lines and projected an image onto the opposite wall. Based on such experimentation, he concluded that vision is accomplished by rays coming from external luminous sources and entering the eye, rather than through rays emitted from the eye as was then commonly believed. He is the first one to experiment with light and found important properties like the rectilinear propagation of light.



Pinhole Camera

Pin hole camera is a simple device which helps us to understand about the rectilinear propagation of light



The above picture shows a model of a pin – hole camera. O is small hole by a pin. XY is the object and Y'X' is the image of XY. As light travels in straight line, one light ray from X travels along the XO strikes the screen X'.

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